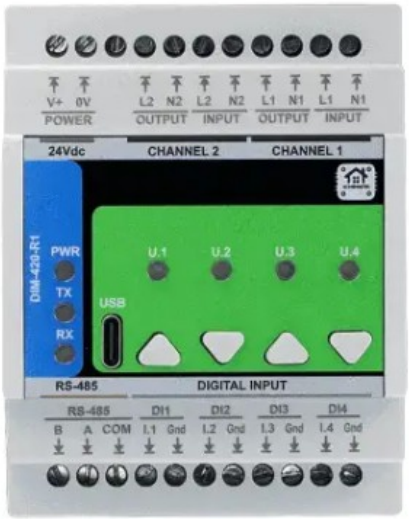
Datasheet

DIM-420-R1 Module

Part No. DIM-420-R1



Manufacturer: ISYSTEMS AUTOMATION S.R.L.
Hardware Version: V1.0
Architecture: Modular (MCU Board + MOSFET Board + Field Board)
Controller: RP2350

Functional Overview

The system is implemented using three hardware boards:

Board	Function
MCU Board	CPU, communication interfaces, USB configuration
Field Board	Digital inputs, signal protection, field interface
MOSFET Board	AC dimming outputs and load switching

The DIM-420-R1 is a modular dimmer I/O module designed for industrial and building automation systems.

It provides 4 IEC 61131-2 compliant 24 V digital inputs (not galvanically isolated) and 2 AC dimming outputs for monitoring field signals and controlling lighting loads using phase-cut dimming.

The module operates from a 24 V DC supply and includes digital input protection, MOSFET-based AC dimming stages, and communication interfaces for integration with PLCs and distributed control systems.

Its modular architecture and DIN-rail form factor make it suitable for control cabinet.

The module communicates with controllers through RS-485 Modbus RTU and can be configured via USB-C interface.

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (W25Q32JV)
Power Input	24 V DC nominal
File system	LittleFS persistent configuration storage
Document revision	DS-DIM-420-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
Digital Inputs	4 × IEC 61131-2 compliant 24 V digital inputs (ISO1212 front-end) (DIx + GNDx), dry-contact (module-wetted), not galvanically isolated; per-channel PTC fuse, TVS surge suppression, reverse-polarity protection, EMI filtering
Dimmer outputs	2 × MOSFET phase-cut dimmer channels (leading / trailing edge, selectable); 1 A RMS per channel continuous, Ta ≤ 40 °C
User Interface	4 buttons, 9 LEDs (power, RX, TX, 4 user; 2 fitted, not assigned)
RS-485	half-duplex Modbus RTU, not galvanically isolated
USB	USB-C (ESD protected, configuration and firmware upload)
Modbus defaults	Address 3, 19200 baud, 8N1

Top Terminal Block (1–10)

Pos	Marking	Signal	Description
1	V+	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	L2	CH2_OUTPUT	Channel 2 dimmed line out (MAINS)
4	N2	CH2_OUTPUT	Channel 2 neutral out (MAINS)
5	L2	CH2_INPUT	Channel 2 line in (MAINS)
6	N2	CH2_INPUT	Channel 2 neutral in (MAINS)
7	L1	CH1_OUTPUT	Channel 1 dimmed line out (MAINS)
8	N1	CH1_OUTPUT	Channel 1 neutral out (MAINS)
9	L1	CH1_INPUT	Channel 1 line in (MAINS)
10	N1	CH1_INPUT	Channel 1 neutral in (MAINS)

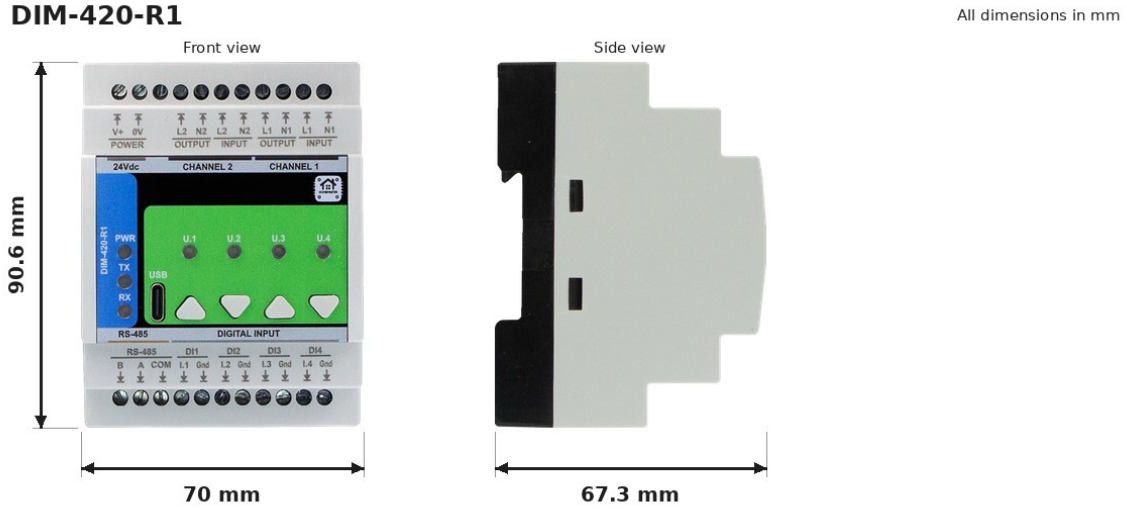
Bottom Terminal Block (1–11)

Pos	Marking	Signal	Description
1	B	RS485	RS-485 data -
2	A	RS485	RS-485 data +
3	COM	RS485	RS-485 signal reference

4	I.1	DI1	DI1 input
5	Gnd	DI1	DI1 return
6	I.2	DI2	DI2 input
7	Gnd	DI2	DI2 return
8	I.3	DI3	DI3 input
9	Gnd	DI3	DI3 return
10	I.4	DI4	DI4 input
11	Gnd	DI4	DI4 return

Dimmer outputs are MOSFET phase-cut channels, 1 A RMS per channel continuous (Ta ≤ 40 °C). Fit an upstream MCB/RCD and a fuse or breaker per channel sized to the load. Mains voltage is present on the load terminals.

Installation, Environmental & Mechanical



Mechanical drawing: front and side view, dimensions in mm

Category	Specification	Details
Environmental Ratings	Operating temperature	0 °C to +40 °C
Environmental Ratings	Storage temperature	−10 °C to +55 °C
Environmental Ratings	Relative humidity	0–90 % RH, non-condensing
Environmental Ratings	Ingress protection	IP20 (inside cabinet only)
Environmental Ratings	Installation	Indoor control cabinet only; not for outdoor or exposed installation
Environmental Ratings	Maximum altitude	2000 m
Environmental Ratings	Pollution degree	2
Mechanical & Packaging	Dimensions	70 × 90.6 × 67.3 mm (L × W × H)
Mechanical & Packaging	DIN width	4 modules (≈ 70 mm)
Mechanical & Packaging	Mounting	35 mm DIN rail
Mechanical & Packaging	Enclosure	PC/ABS industrial enclosure
Mechanical & Packaging	Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Mechanical & Packaging	Wire cross-section	0.2–2.5 mm² (AWG 24–12)
Mechanical & Packaging	Tightening torque	0.4–0.6 Nm

Mechanical & Packaging	Net weight	210 g
Mechanical & Packaging	Gross weight	320 g
Mechanical & Packaging	Pack size	TBD

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.

Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

RS-485 Cable

- **Construction:** twisted pair for **A/B**
- **Characteristic impedance: 120 Ω** (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples: J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**
- Use one twisted pair for A/B. Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Shield Grounding

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

Compliance & Certifications

The DIM-420-R1 is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

EU Directives

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Harmonised Standards

Area	Standards
EMC	EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions)
Electrical Safety	EN 62368-1
RoHS	EN IEC 63000

Safety notice

Dimmer outputs are MOSFET phase-cut channels, 1 A RMS per channel continuous (Ta ≤ 40 °C). Fit an upstream MCB/RCD and a fuse or breaker per channel sized to the load. Mains voltage is present on the load terminals.

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